Lecture Notes 1-2, Probability

Any experiment involving randomness can be modelled as a probability space. Such a space comprises a set Ω of possible outcomes of the experiment, a set $\mathcal{F}$ of events, and a probability measure $\mathbb{P}$. The definition and basic properties of a probability space are explored, and the concepts of conditional probability and independence are introduced. Many examples involving modelling and calculation are included.

Let's introduce the notion of probability by the example of tossing two coins (each showing "head": H or "tail": T). The possible outcomes in this example are:

$$HH, HT, TH, TT.$$

All of these outcomes are equally likely, and the sum of the probabilities is 100%, that is 1, so each elementary event here has probability $\frac{1}{4}$. (Don't get confused: HT and TH are different outcomes! Try if you don't believe: in the long run, having one tail and one head is twice as frequent as having two heads. Even if the coins look the same, they can be identified. If you have doubt about the probability of an event because of symmetry like in this case, identifying the participants always gives true result.)

We denote *the set of all possible outcomes* by Ω:

$$\Omega = \{HH, HT, TH, TT\}. \qquad \mathbb{P}(\Omega) = 1.$$

The simple or atomic or *elementary events* are: $\{HH\}, \{HT\}, \{TH\}, \{TT\}$.

A non-elementary *event* is: "exactly one coin is showing head"$= \{HT, TH\}$.

Events are subsets of Ω. We can denote events by capital letters $(A, B, ...)$. Further events: "the first coin is showing head"$= \{HH, HT\} = A$, or "the second coin is showing head"$= \{HH, TH\} = B$. Their intersection is $A \cap B = \{HH\}$. Their union is $A \cup B = \{HH, HT, TH\}$.

De Morgan's laws about union and intersection of events are:

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c,$$

where c means the *complement* (negation) of an event.

A basic rule about probabilities:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Now let's discuss an *example* (for simplicity, I'm going to use P for probability instead of $\mathbb{P}$). After rolling two dice, let X be the number of 6's shown, and let Y be the number of even numbers shown on them.

$$P(X=0) = \frac{25}{36}, \; P(X=1) = \frac{10}{36}, \; P(X=2) = \frac{1}{36},$$
$$P(Y=0) = \frac{9}{36}, \; P(Y=1) = \frac{18}{36}, \; P(Y=2) = \frac{9}{36}.$$

Think it over!

X and Y are called *random variables.* Convention: capital letters refer to random variables, that can have multiple different outcomes. (Small letters will refer to real numbers: e.g. small x can be used for the actual outcome of the random variable X).

Joint distribution of X and Y:

	X=0	X=1	X=2	sum
Y=0	$\frac{9}{36}$	0	0	$\frac{9}{36}$
Y=1	$\frac{12}{36}$	$\frac{6}{36}$	0	$\frac{18}{36}$
Y=2	$\frac{4}{36}$	$\frac{4}{36}$	$\frac{1}{36}$	$\frac{9}{36}$
sum	$\frac{25}{36}$	$\frac{10}{36}$	$\frac{1}{36}$	$\frac{36}{36}$

Note that the sums of the probabilities in columns agree with the distribution of X, and the sums of the probabilities in rows agree with the distribution of Y given above. These summed probabilities build up the *marginal distributions* in this context.

The above table of joint probabilities is called a *contingency table* or *cross-tab*.

Intuitively it is obvious that X and Y are not independent (because 6 is an even number). Mathematical definition of *independence of events* is a consequence of the notion of conditional probability, to be discussed later, but here I give the definition in advance.

Events A and B are called *independent* if $P(A \cap B) = P(A)P(B)$.

More generally: a family $\{A_i : i \in I\}$ is called independent if $P(\bigcap_{i \in J} A_i) = \prod_{i \in J} P(A_i)$, for any $J \subseteq I$ set of indices.

Pairwise independence: when $P(A_i \cap A_j) = P(A_i)P(A_j)$ holds for any A_i and A_j, $i \neq j$. It does not imply independence!

X and Y (two random variables) are called independent if the above definition holds for the sets $\{X \leq k\}$ and $\{Y \leq l\}$ for any k, l real numbers. We can define independence similarly for a family of random variables.

Example 1: Let's check the definition of independence on the random variables X and Y in the above contingency table. They are not independent indeed, a counterexample is: $P(X = 0 \text{ and } Y = 0) = \frac{9}{36}$, $P(X = 0)P(Y = 0) = \frac{9}{36} \times \frac{25}{36}$. These two expressions are clearly not equal.

Example 2: Consider the experiment of rolling two dice again, but now let X mean the number shown on the first die, and Y mean the number shown on the second die. Let Z denote their sum modulo 6 (that is, the remainder after division by 6). E.g. when X=2 and Y=5 then $Z \equiv 2 + 5 = 7 \equiv 1 \pmod 6$. The possible values of Z are $0, 1, 2, 3, 4, 5$ (as we will see).

X and Y are obviously independent. X and Z are also independent because e.g. when $X = 3$ and Y takes the values $1, 2, 3, 4, 5, 6$, then the values of Z are $4, 5, 0, 1, 2, 3$, respectively. We get the same set for any value of X, each outcome with probability $\frac{1}{6}$. Similarly, Z is independent of Y. But X, Y and Z are not independent as a family. E.g. $P(X = 1, Y = 2, Z = 4)$ is zero, because 1+2 (mod 6) is not 4. This is not equal to the product of the individual probabilities: $P(X = 1)P(Y = 2)P(Z = 4) = \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \frac{1}{216}$.

In the above examples, we had a sequence of independent experiments: rolling dice independently of each other. If we do not have dice, we can simulate the same experiment by having six numbered objects in a box, taking one out, *replacing* it in the box, then taking out another object. This is one major type of combinatorial exercise.

The other major type of combinatorial exercise is when we choose some elements from a set, *without replacement.*

Notation: $\binom{n}{k}$ = the number of k-element subsets of an n-element set, called *"n choose k"*:

$$\binom{n}{k} = \frac{n(n-1)...(n-k+1)}{k!} = \frac{n!}{k!(n-k)!} = \binom{n}{n-k}$$

Example: Pick 5 cards randomly out of a pack of 32 cards (Hungarian card, with 4 different colours and 8 different figures). Which event has bigger probability: having 5 cards of identical colour, or having 4 cards of identical figure among the 5 cards?

Solution: $P(\text{choosing 5 cards of identical colour}) = \frac{4 \times (8 \times 7 \times 6 \times 5 \times 4)}{32 \times 31 \times 30 \times 29 \times 28} = \frac{4\binom{8}{5}}{\binom{32}{5}} = \frac{1}{29 \times 31}$, and

$P(\text{having 4 cards of identical figure among 5 chosen cards}) = \frac{8 \times \binom{4}{4} \times 28}{\binom{32}{5}} = \frac{1}{29 \times 31}$, so they are equally likely!

Many statements about chance take the form "if B occurs, then the probability of A is p", where B and A are events (such as "it rains tomorrow" and "the bus being on time" respectively) and p is a likelihood. To include this in our theory, we return briefly to the discussion about proportions. An experiment is repeated N times, and on each occasion we observe the occurrences or non-occurrences of two events A and B. Now, suppose we only take an interest in those outcomes for which B occurs; all other experiments are disregarded. (That means, "our world" consists of those elementary events that belong to B.) In this smaller collection of trials the proportion of times that A occurs is $\frac{N(A\cap B)}{N(B)}$, since B occurs at each of them. However,

$$\frac{N(A\cap B)}{N(B)} = \frac{N(A\cap B)/N}{N(B)/N}.$$

If we now think of these ratios as probabilities, we see that the probability that A occurs, given that B occurs, should be reasonably defined as $P(A\cap B)/P(B)$.

Conditional probability (probability of A, if B occurs) is denoted by $P(A|B)$. Based on the above argument, it is defined as

$$P(A|B) = \frac{P(A\cap B)}{P(B)}.$$

In case of independence, the occurrence of A does not depend on the occurrence of B, so $P(A|B)$ must be equal to $P(A)$. This yields the definition of independence given earlier.

Consider our first *example* with two dice again, X denoting the number of 6's and Y denoting the number of even numbers shown on the dice. The definition of conditional probability yields e.g. $P(X=2|Y=0)=0$, and $P(X=1|Y=1) = \frac{6/36}{18/36} = \frac{1}{3}$. The first one is obvious intuitively (we can't have two sixes if we have no even numbers). The second one can be seen intuitively as well: out of $2,4,6$ (possible even numbers on a die) only 6 is suitable for $X=1$ (and it doesn't matter which odd number we have on the other die; also doesn't matter which die shows the even number).

To deduce a formula for *reversing* conditional probabilities (find $P(B_i|A)$ when $P(A|B_i)$ is given, see exact conditions later), we use the equation $P(A|B)P(B) = P(A\cap B)$ that is equivalent to the definition of conditional probability. We apply this expression in the following *example*.

Only two factories manufacture zoggles. 20 per cent of the zoggles from factory I and 5 per cent of the zoggles from factory II are defective. Factory I produces twice as many zoggles as factory II each week. What is the probability that a zoggle, randomly chosen from a week's production, is satisfactory? Clearly this satisfaction depends on the factory of origin. Let A be the event that a chosen zoggle is satisfactory, and let B the event that it was made in factory I.

$$P(A) = P(A\cap B) + P(A\cap B^c) = P(A|B)P(B) + P(A|B^c)P(B^c) = 0.8\times\frac{2}{3} + 0.95\times\frac{1}{3} = \frac{51}{60}.$$

If the chosen zoggle is defective, what's the probability that it came from factory I? In our notation, this is just $P(B|A^c)$:

$$P(B|A^c) = \frac{P(B\cap A^c)}{P(A^c)} = \frac{P(A^c|B)P(B)}{P(A^c)} = \frac{0.2\times\frac{2}{3}}{1-\frac{51}{60}} = \frac{8}{9}.$$

A formal statement, based on a similar argument (with n sets instead of 2) is *Bayes's formula*: if $B_1, B_2, ..., B_n$ is a partition of Ω (that means they are disjoint and their union is Ω), where each B_i has a positive probability, then

$$P(B_j|A) = \frac{P(B_j\cap A)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^{n} P(A|B_i)P(B_i)}.$$

There is one more concept I want to discuss here in the beginning, even thought it is not part of the first example sheet. It is *the expectation of a random variable* (also called *expected value* or *mean*, the latter word referring to the *average*, too).

Let us consider *the expectation of a game* first, on 3 examples.

1) In the first game, you can win 1 dollar with probability $\frac{1}{2}$ (50%), or you can loose 1 dollar, with probability $\frac{1}{2}$ (50%) as well. The expectation of this game is zero, because in the long run, your wins and losses tend to be equal. Mathematically, if your gain is X, its expectation is $E(X) = \frac{1}{2} \times 1 + \frac{1}{2} \times (-1) = \frac{1}{2} - \frac{1}{2} = 0$. This is a theoretical value, real outcomes may differ, but you can see the symmetry of the game that must convince you about this expectation. Games with zero expectation are called *fair* games.

2) In the second game, you can win 1 dollar with probability 95%, but you can loose 20 dollars with probability 5%. Is this game fair as well? - Almost, but not quite. The expected value is $E(X) = 0.95 * 1 - 0.05 * 20 = 0.95 - 1 = -0.05$. It is negative, so in the long run, one tends to lose on this game. The rule for calculating the expected value is: summing over all possible outcomes, each outcome (value) multiplied by its probability. This is good because the probability shows the expected relative frequency. This is a weighted average, where the weights are the probabilities of the different outcomes.

3) In Roulette, a wide-spread strategy is to put 1 dollar on Red, which wins with probability 18/37. If you win, you get 2 dollars in place of your 1 dollar. If you loose this stake, you put 2 dollars on Red in the second round. Then, you either win 4 in place of 2, or you just loose your 2 dollars. And so on: you only stop when you first win, but until then, you always double your stake on Red. If you first win in the nth turn, you have put on $1 + 2 + 4 + ... + 2^{n-1}$, but now you get back $2 * 2^{n-1} = 2^n$, which means you own 1 dollar more then before going through this sequence. You can only loose if you don't have enough money to double your stake (which is possible).

Let's calculate what are your chances after the first 3 turns.

a) You can win +1 immediately with probability $\frac{18}{37}$.

b) You can loose first with probability $\frac{19}{37}$, then win in the second turn with probability $\frac{18}{37}$. The value of this outcome is +1 dollar.

c) You can loose twice with probability $(\frac{19}{37})^2$, then win in the third turn with probability $\frac{18}{37}$; the value of this outcome is again +1 dollar.

d) You can loose three times with probability $(\frac{19}{37})^3$, and if you stop here, you lost $1 + 2 + 4 = 7$ dollars.

So with this strategy, your expectation for 3 rounds of the game is:

$$E(X) = \frac{18}{37} \times 1 + \frac{19}{37} \cdot \frac{18}{37} \times 1 + \left(\frac{19}{37}\right)^2 \cdot \frac{18}{37} \times 1 + \left(\frac{19}{37}\right)^3 \times (-7) = -0,08329.$$

The expectation is slightly negative: again, there is a tendency to loose on this game.

Later on, we will study the notion of expectation further. We can calculate the expected value of random variables in general. Let's practice it, going back to the first *example (with the contingency-table):*

$E(X) = \frac{25}{36} \times 0 + \frac{10}{36} \times 1 + \frac{1}{36} \times 2 = \frac{12}{36} = \frac{1}{3}$,

$E(Y) = \frac{9}{36} \times 0 + \frac{18}{36} \times 1 + \frac{9}{36} \times 2 = \frac{36}{36} = 1$,

$E(XY) = ... \times 0 + \frac{6}{36} \times 1 + (\frac{4}{36} + 0) \times 2 + \frac{1}{36} \times 4 = \frac{6+8+4}{36} = \frac{1}{2}$.

There is a frequently used notion that we can easily calculate now, called *covariance*. It is in connection with independence: if X and Y are independent, their covariance is zero. The reverse is not true: zero covariance does not imply independence. The definition of covariance is: $cov(X, Y) = E(XY) - E(X)E(Y)$.

Here $cov(X, Y) = \frac{1}{2} - \frac{1}{3} \times 1 = \frac{1}{6}$, not zero. Indeed, X and Y are not independent.